Customer Behaviour Analysis

Uncovering what drives purchases — from demographics to shipping preferences — to power smarter business decisions.

Made with GAMMA

Dataset at a Glance

3,900

Rows

18

Columns

37

Missing Values

Review rating column

Demographics

Age, Gender, Location, Subscription

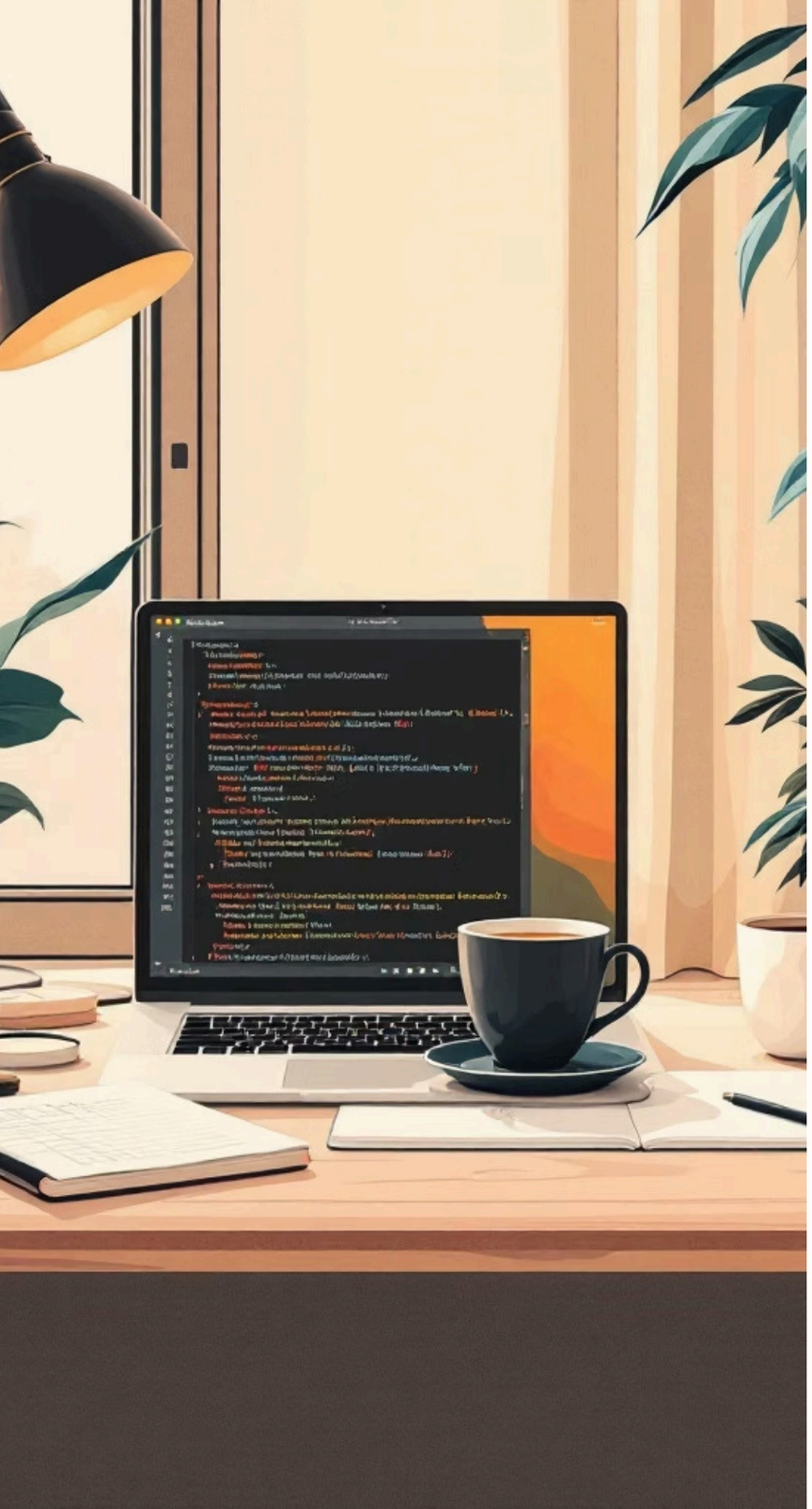
Purchase Details

Item, Category, Amount, Season, Size, Colour

Shopping Behaviour

Discounts, Promo codes, Frequency, Ratings, Shipping

Made with GAMMA



Exploratory Data Analysis in Python

01

Load & Explore

Imported dataset via pandas; reviewed summary statistics

02

Clean & Standardise

Filled missing ratings with category median; renamed columns to snake_case

03

Feature Engineering

Created **age_group** column; converted purchase frequency to **days between purchases**

04

Consistency & Integration

Dropped redundant **promo_code_used**; connected to PostgreSQL

Made with GAMMA

SQL Analysis: Key Business Questions

Approach

After Python preprocessing, data was connected to **PostgreSQL** to answer 10 targeted business questions.

Revenue

By gender & age group

Behaviour

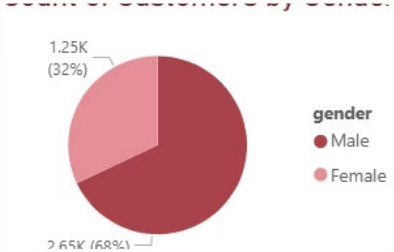
Discounts, shipping, subscriptions

Products

Ratings, categories, top sellers

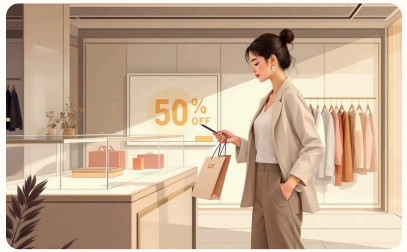
Made with GAMMA

Revenue & Spending Insights



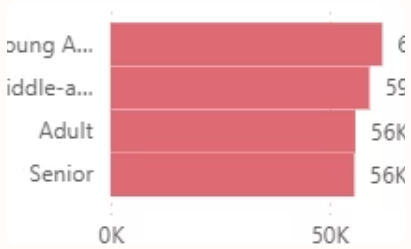
Customers by Gender

Compared total count — male vs. female customers



High Spending Discounters

Customers using discounts yet spending **above average**



Revenue by Age Group

Total revenue calculated per age segment

Made with GAMMA

Product & Shipping Analysis

Top 5 by Rating

Products with highest average review ratings

Top 3 per Category

Most purchased products within each category

Discount-Dependent

5 products with highest % of discounted purchases

Shipping Type Comparison

Average purchase amounts compared between **standard** and **express** shipping. Purchase amounts are higher with **express** shipping

Made with GAMMA

Customer Segmentation

New - 83

First-time buyers

Returning - 701

Repeat purchasers

Loyal - 3116

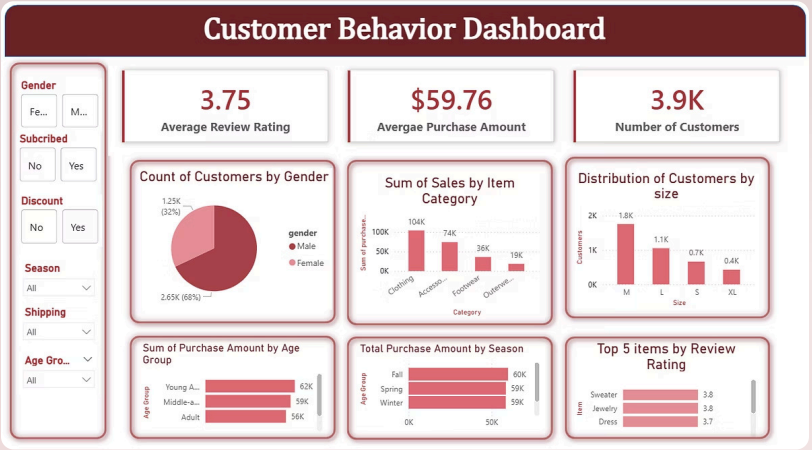
High-frequency, long-term customers

Customers with **more than 5 purchases** were also analysed for subscription likelihood — revealing a strong link between repeat buying and subscription status.

Made with GAMMA

Power BI Dashboard

All findings visualised in an interactive Power BI dashboard — enabling dynamic exploration of customer behaviour, revenue trends, and product performance.



Made with GAMMA

Business Recommendations

Boost Subscriptions

Add subscriber-only perks to convert non-subscribers

Loyalty Program

Push new & returning customers toward repeat purchases

Targeted Marketing

Focus on middle-age & young adults, spring items

Smart Inventory

Stock more male, medium-size & clothing items

Product Positioning

Highlight top-rated & best-selling products in campaigns

Made with GAMMA

Key Takeaways

Data-driven decisions

3,900 records cleaned, engineered, and analysed end-to-end

Full pipeline

Python → PostgreSQL → Power BI

Actionable insights

5 clear recommendations to grow revenue and loyalty

Made with GAMMA